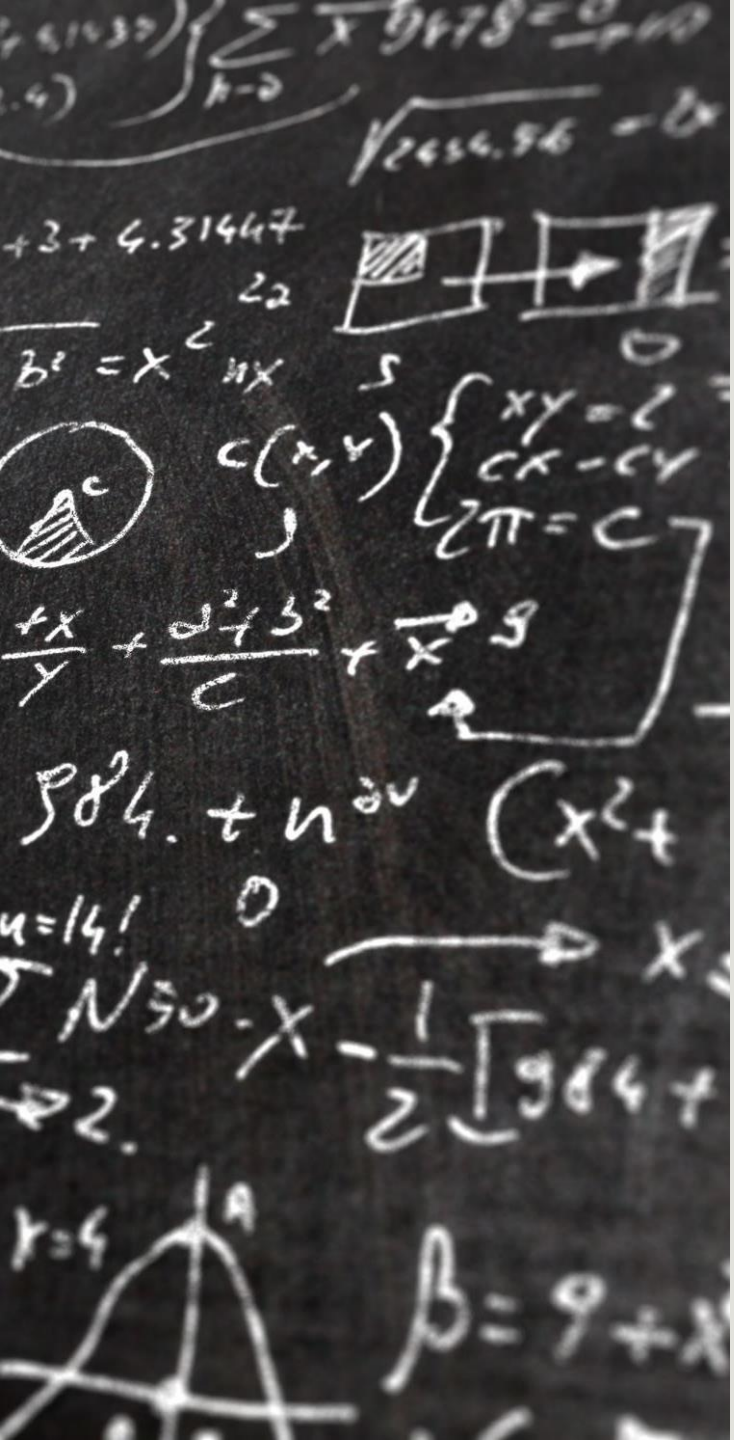


Understanding UML Class Diagrams & Requirements Traceability

MANAGING CHANGES AND ENSURING PROJECT SUCCESS

Introduction to Classes

- A class is a blueprint for creating objects.
- Classes encapsulate state (attributes) and behavior (methods).

Example:

Class: Dog

Attributes: name, color, breed

Methods: bark(), wagTail(), eat()

UML Class Diagram Basics

- Three parts of a class diagram box:

  1. Class Name
  2. Attributes
  3. Methods (Operations)

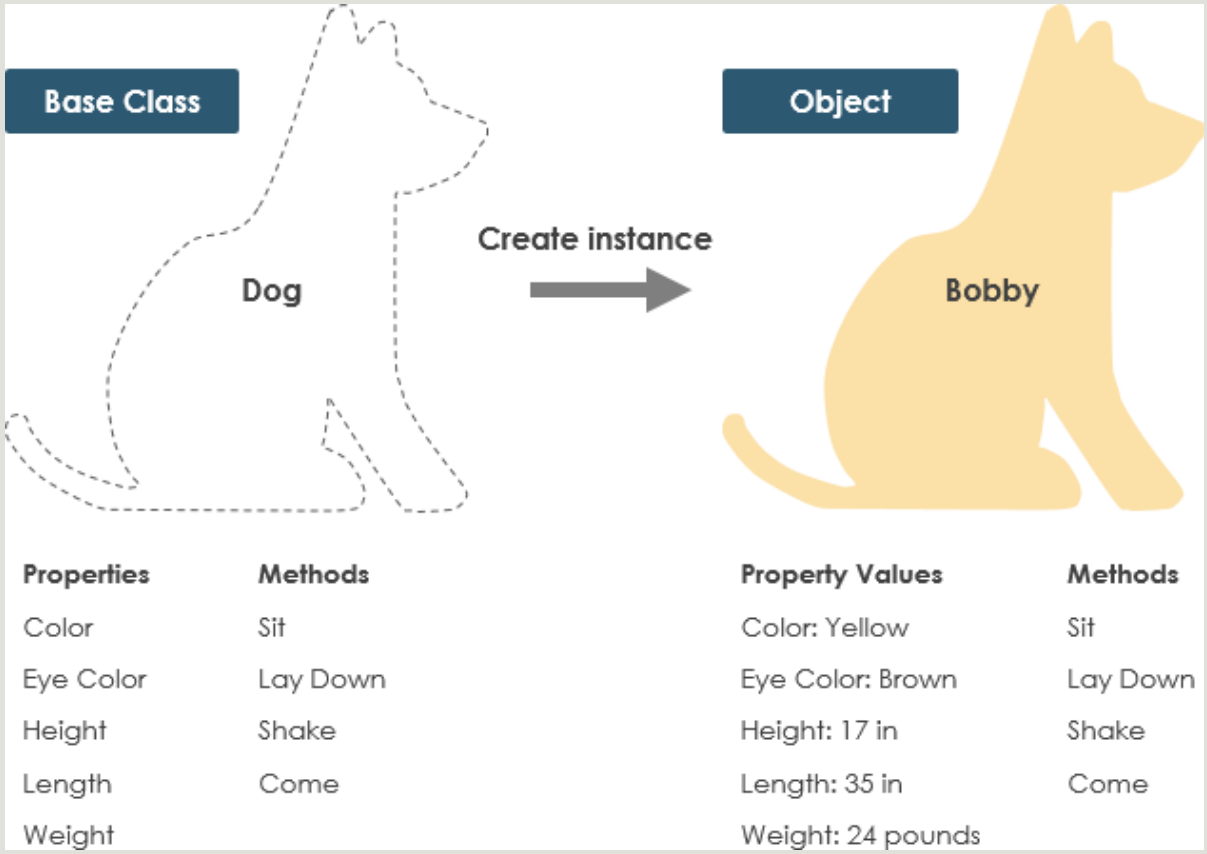
- Visibility Modifiers:

  + Public

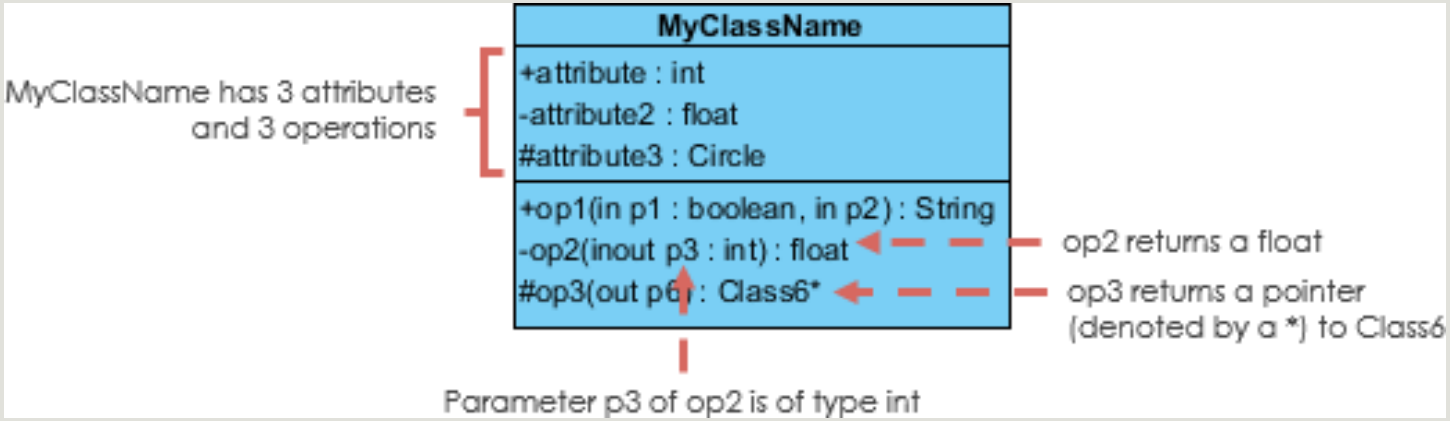
  - Private

  # Protected

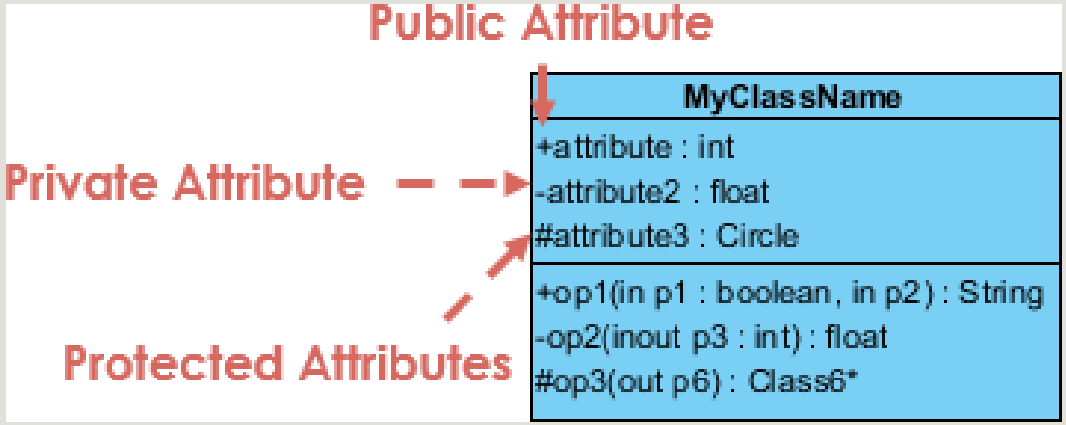
UML Class Diagram Basics



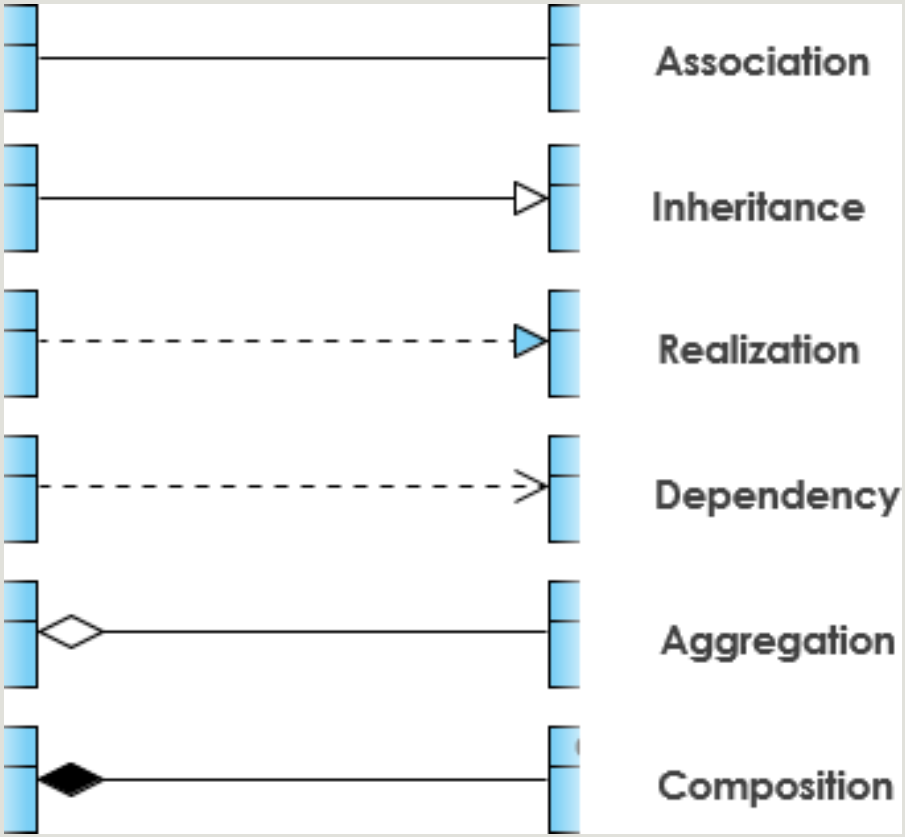
UML Class Diagram Basics



UML Class Diagram Basics

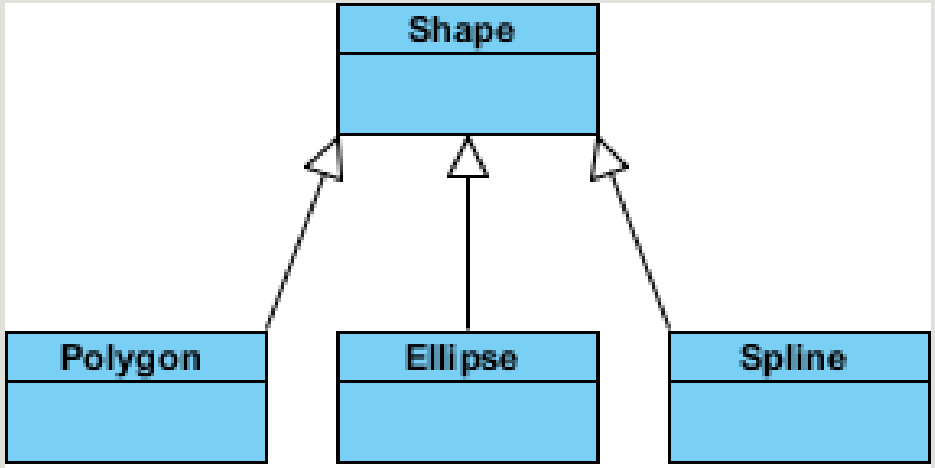


Relationships Between Classes

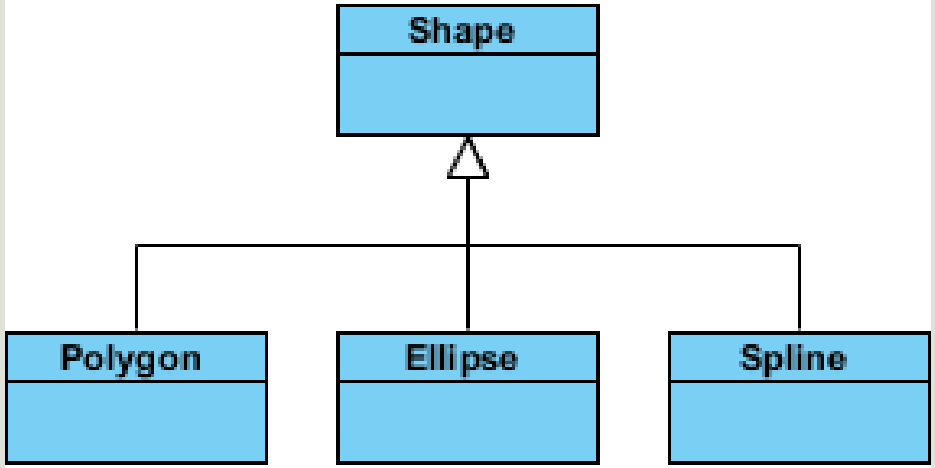


Generalization (Inheritance)

- Definition: 'Is-a' relationship.
- Example:
  - Superclass: Animal
  - Subclass: Dog, Cat
  - A Dog is an Animal.

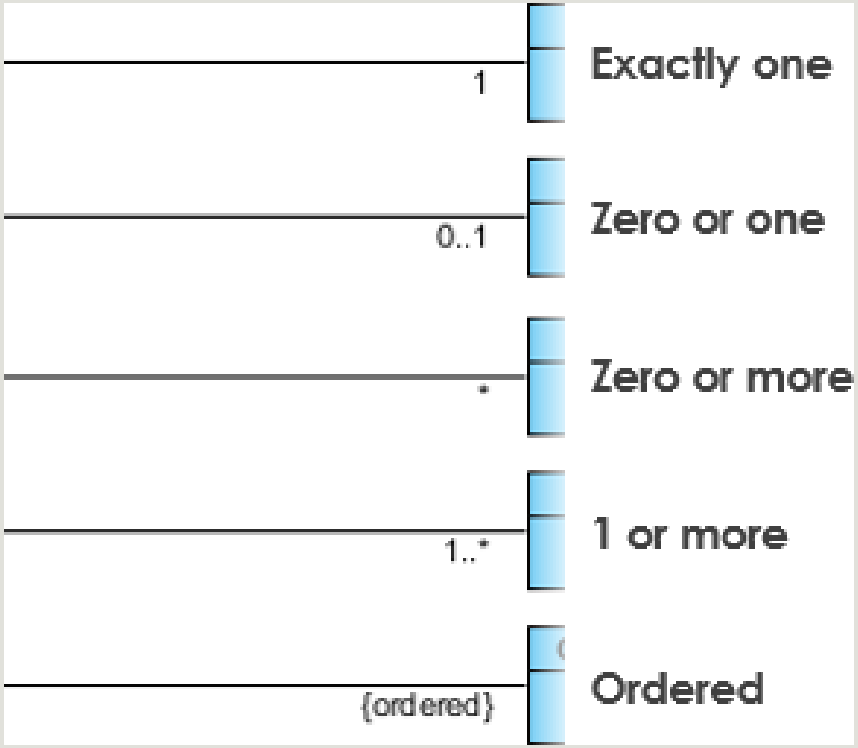


Style 1: Separate target

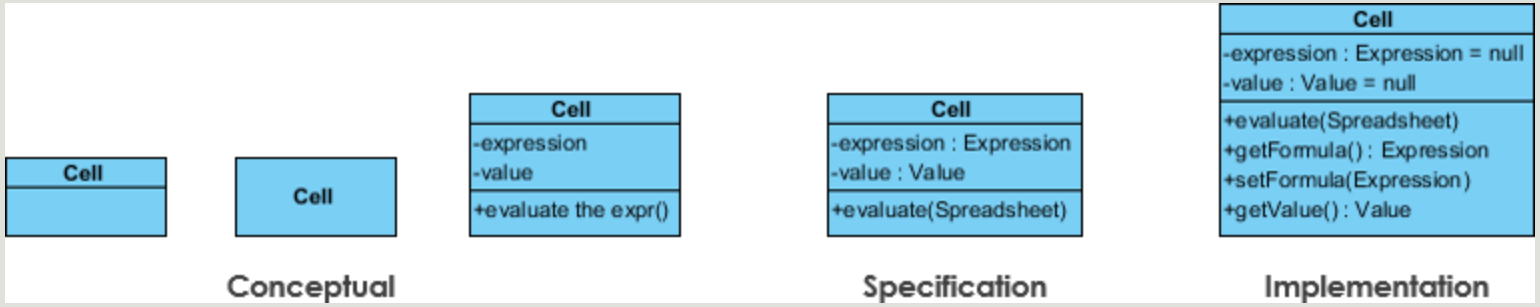


Style 2: Shared target

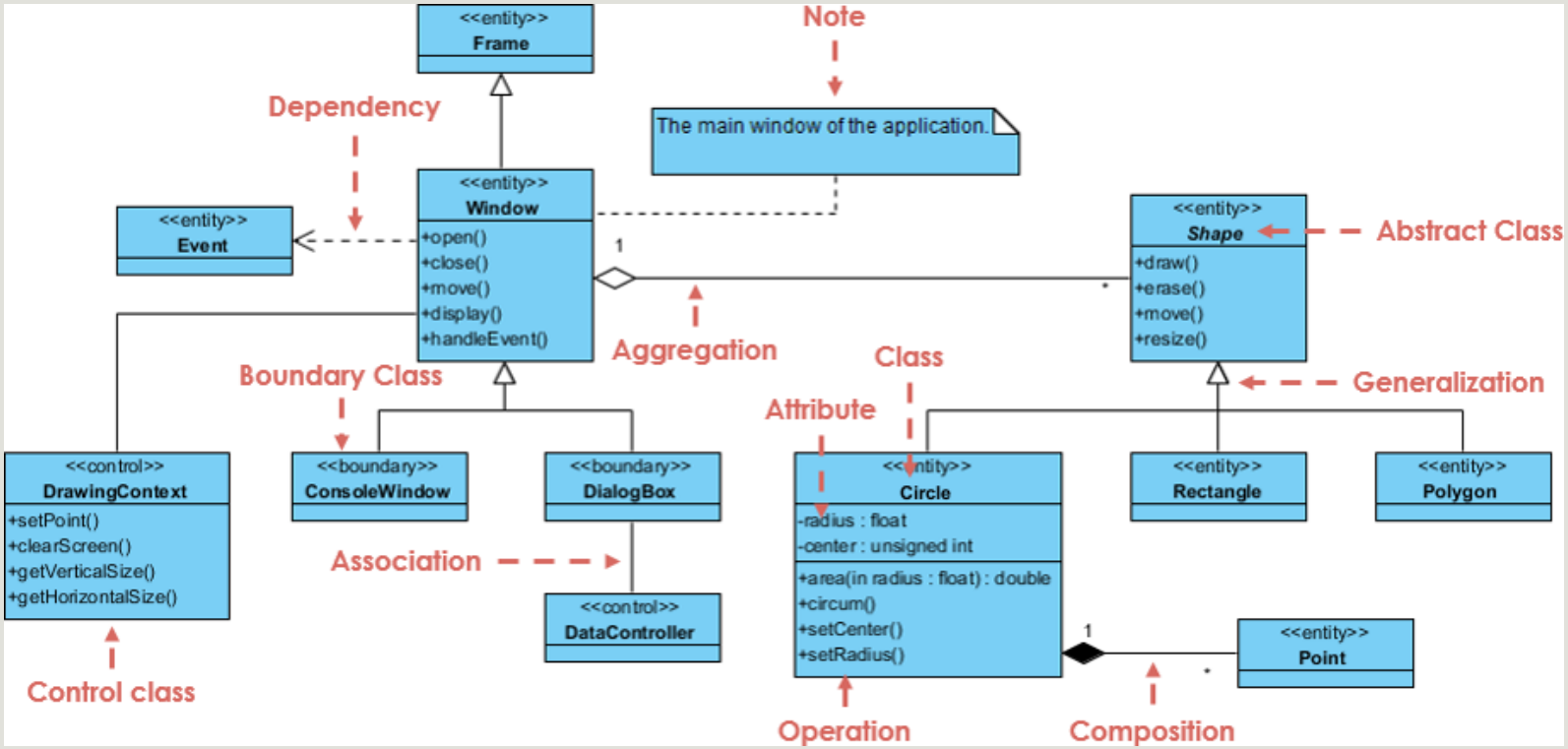
Cardinality



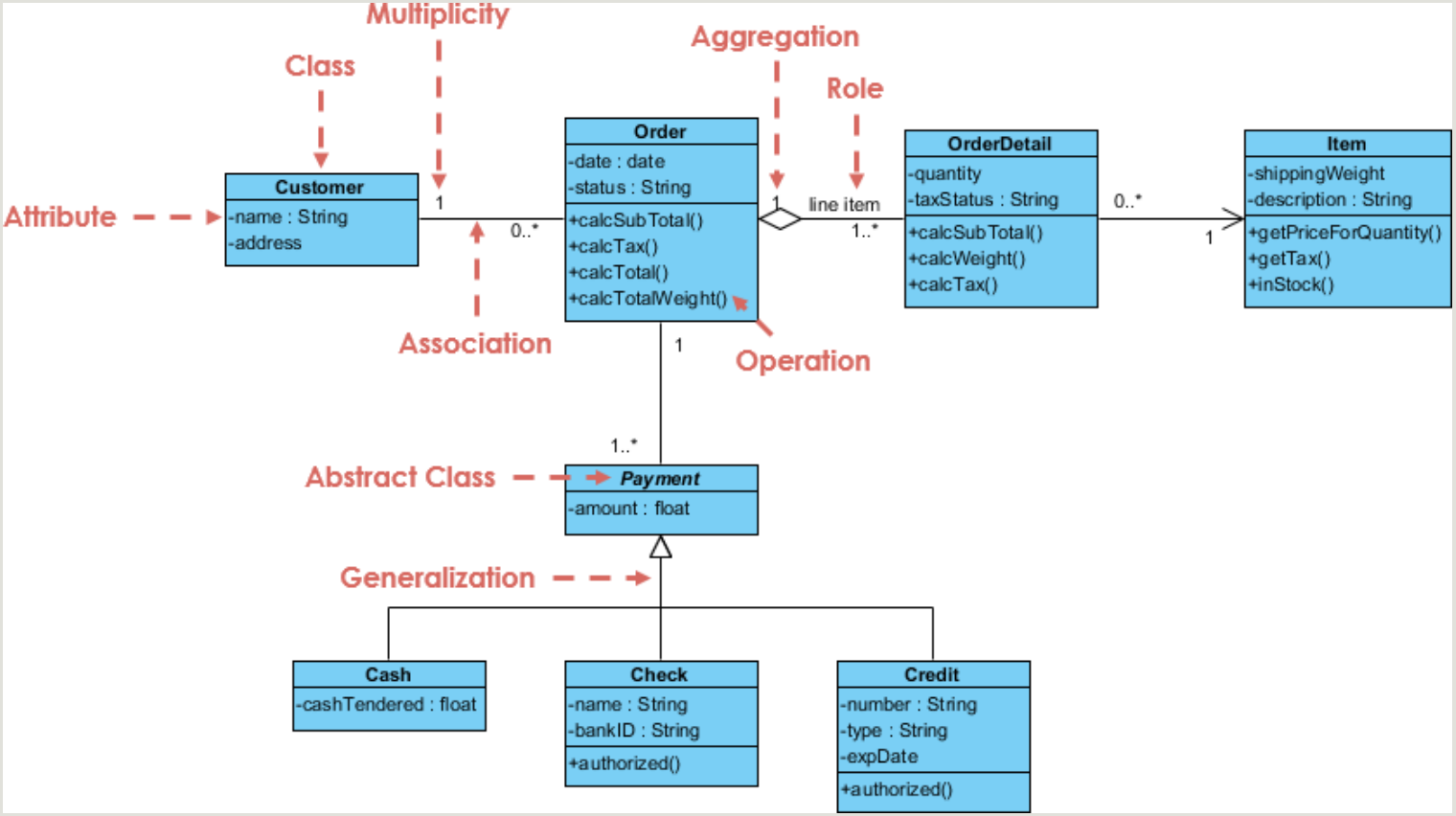
Perspectives of Class Diagrams



Example


Note
<<entity>>
Frame
Dependency
The main window of the application.
<<entity>>
Window
+open()
+close()
+move()
+display()
+handleEvent()
<<entity>>
Event
1
<<entity>>
Shape
+draw()
+erase()
+move()
+resize()
Abstract Class
Aggregation
Class
Boundary Class
Generalization
Attribute
<<control>>
DrawingContext
+setPoint()
+clearScreen()
+getVerticalSize()
+getHorizontalSize()
<<boundary>>
ConsoleWindow
<<boundary>>
DialogBox
<<entity>>
Circle
-radius : float
-center : unsigned int
+area(in radius : float) : double
+circum()
+setCenter()
+setRadius()
<<entity>>
Rectangle
<<entity>>
Polygon
Association
<<control>>
DataController
1
<<entity>>
Point
Control class
Operation
Composition

Example


Multiplicity
Class
Aggregation
Role
Attribute
Association
Operation
Abstract Class
Generalization
Customer
-name : String
-address
Order
-date : date
-status : String
+calcSubTotal()
+calcTax()
+calcTotal()
+calcTotalWeight()
OrderDetail
-quantity
-taxStatus : String
+calcSubTotal()
+calcWeight()
+calcTax()
Item
-shippingWeight
-description : String
+getPriceForQuantity()
+getTax()
+inStock()
line item
1
0..*
1..*
Payment
-amount : float
Cash
-cashTendered : float
Check
-name : String
-bankID : String
+authorized()
Credit
-number : String
-type : String
-expDate
+authorized()

Requirements Traceability

- Definition: Traceability ensures every requirement is linked to its source and implementation artifacts.
- Purpose:
  - Tracks how requirements influence system design and testing.
  - Provides accountability and ensures nothing is overlooked.

Traceability Matrix Example

Requirements Traceability Matrix

	Testing								Defects			
WBS Deliverables	Test Case ID	Test Description	TEST	UAT	QA	PROD	PRE-PROD	NON-PROD	Defective	Defect ID	Defect Description	Req. Status
WBS-001	TC001	Verify user can successfully register an account	Pass	Pass	Pass	Pass	Fail		No			Complete
WBS-001	TC002	Verify error message is displayed for invalid inputs	N/A	Fail	Pass	Fail	N/A		Yes	DEF001	Invalid email format	Complete
WBS-001	TC003	Verify user receives a confirmation email	Pass	N/A	Fail	N/A	Pass		Yes	DEF002	Email not sent	In Progress
WBS-002	TC004	Verify user can log in with valid credentials	Fail	Pass	N/A	N/A	N/A		NO			In Progress
WBS-002	TC005	Verify error message is displayed for incorrect login	Fail	Fail	N/A	Pass	Fail		Yes	DEF003	Incorrect username	Complete
WBS-002	TC006	Verify "Forgot Password" link redirects correctly	Fail	Fail	N/A	Pass	Pass		No			In Progress
WBS-003	TC007	Verify user can create a new post				N/A	N/A		No			Not Started
WBS-003	TC008	Verify error message is displayed for empty content	Pass	Pass	Pass	Fail	Fail		Yes	DEF004	Empty post content	Complete

Change Management Process

1. Change Request: Document the change.
2. Impact Analysis: Assess effects on scope, cost, timeline.
3. Approval/Denial: Stakeholders decide whether to proceed.
4. Implementation and Verification: Validate the changes.

Case Study

- Classes: Library, Book, Member, Staff.
- Relationships:

  - Library contains many Books (Aggregation).

  - Member borrows Books (Association).

  - Staff manages the Library (Association).
- Simple Traceability Matrix + Change Management Plan

Requirement ID	Description	Design Element	Test Case ID	Status
LIB-01	Search books by title/author	Search Page Design	TC-001	Completed
LIB-02	Overdue notifications	Notification Module	TC-002	In Progress

Wrap-Up and Q&A

- Traceability ensures all requirements are linked to implementation and testing.
- Structured change management adapts to evolving requirements.

Questions:

- What happens if traceability is skipped in a project?
- How would you handle frequent stakeholder changes?